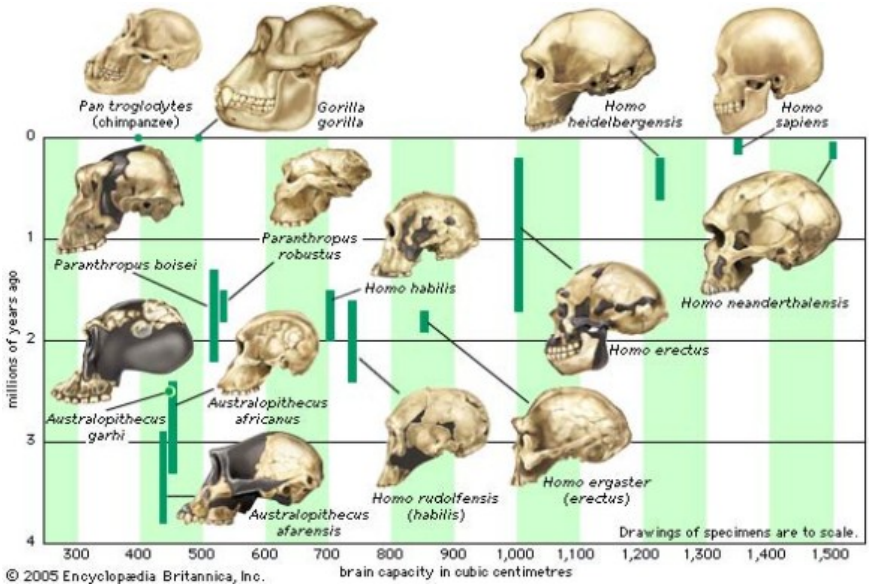
Primate Evolutionary Trends



Digits
Dentition
Vision
Cerebral Cortex
Gestation and Parental Care

Trends

- Primates show a progression in changing characteristics, suggesting an evolutionary trend.
- This can be seen in:
  - Digits
  - Dentition
  - Vision
  - Cerebral Cortex size
  - Gestation and Parental Care.

Digits

- Limbs unspecialised in structure
  - Allows diversity in use
- 5 digits on each limb (pentadactyl)
  - Highly mobile
  - Prehensile
    - can curl around things to grip
  - Opposable thumbs (and big toes for some)
    - Thumb can cross palm to touch other fingers





Digits – degree of opposability

- Varies between species
- Depends on relative length of first digit
- Most primates also have opposable big toe
  - Humans are an exception
  - Feature a disadvantage when bipedal – weight distribution
- Humans have the longest thumb
  - Allows manipulation of materials.

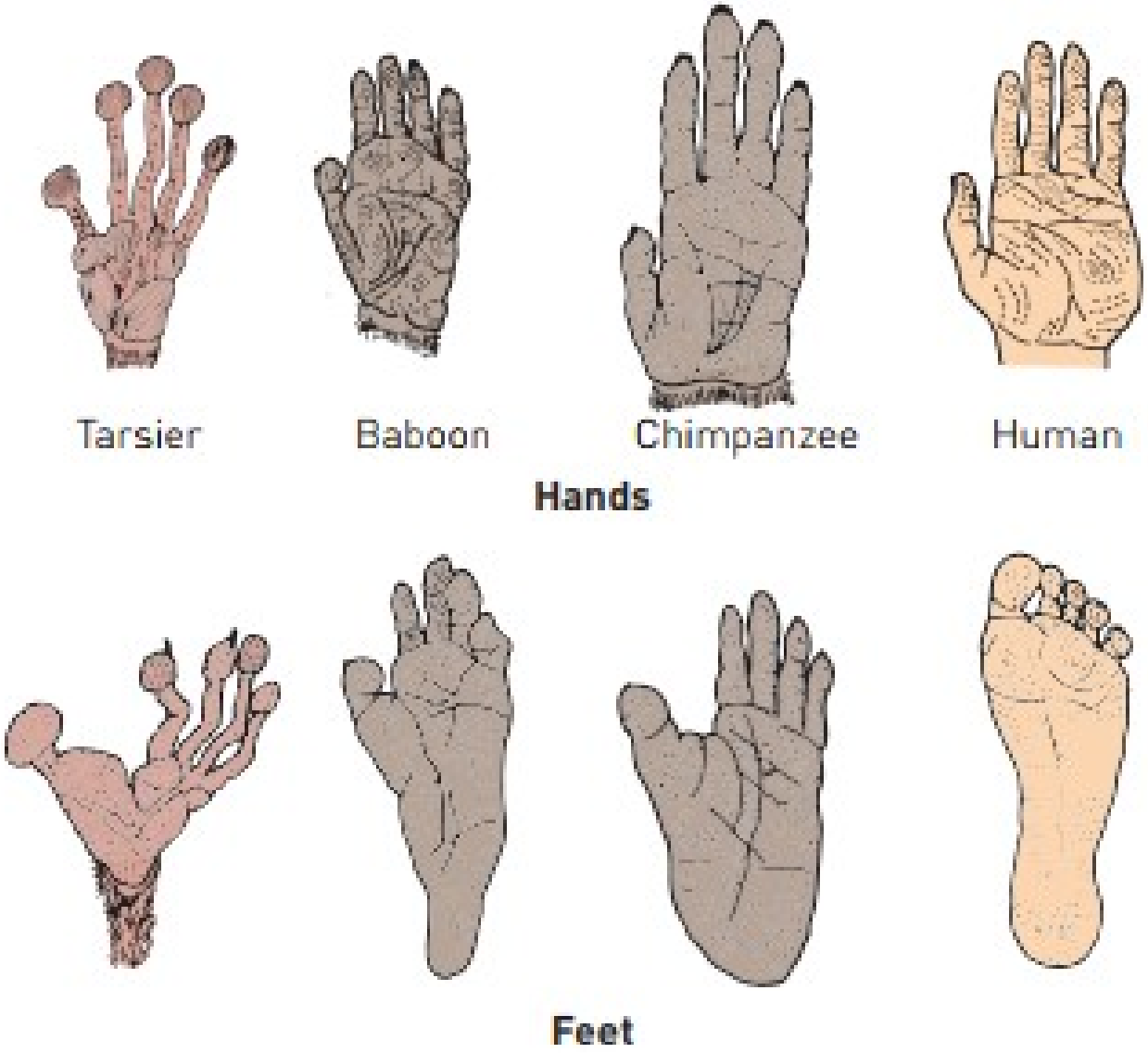


Figure 17.4 Hands and feet of four primates. Unlike other primates, humans do not have an opposable big toe.

Digits – nails vs claws

- Tips of digits modified for better grip in trees
  - Nails instead of claws
  - Nails allow better grasping
  - Evolved from claws that became flattened
- Some primates still have claws – Aye-Ayes and Lemurs
  - Lemurs: “toilet claw” for grooming/scratching
  - Aye Ayes -
- Digits also have more sense receptors for feedback when gripping and manipulating
- Friction ridges “fingerprints” increase grip



Figure 17.8 The fingerprint patterns of a variety of primates

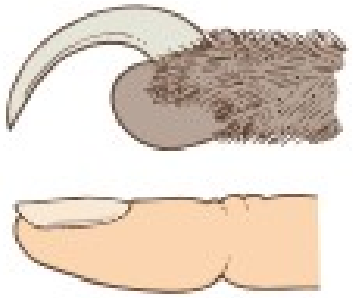


Figure 17.6 Differences between claws and nails

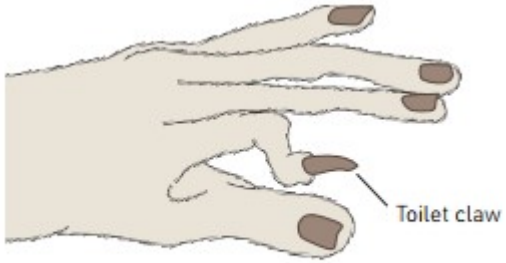


Figure 17.7 The toilet claw of a lemur









Digits – mobility

- Humans – very mobile digits
  - Short broad hand, short straight fingers, long strong thumb
  - Allows precision grip
    - Writing
    - Sewing
    - Construction
- True opposable thumb
  - Only in Old World monkeys eg baboons, mandrills, macaques; apes; humans

Dentition (Teeth)

- Evolutionary changes can be seen to teeth
- Can look at Dental Formula
  - Gives number of each type of tooth on each jaw quarter
  - Primitive mammals: 3:1:4:3 (incisors: canines:premolars:molars)
- Decrease in number of teeth in primates compared with early mammals.
  - Probably related to gradual reduction in face and jaw size
  - Lemurs and lorises: 36 teeth 2:1:3:3
  - Tarsiers: have lost 2 incisors from lower jaw: 2:1:3:3/1:1:3:3
  - New World Monkeys – evolutionary trend for missing third molar
  - Old World Monkeys, Apes and Humans: 32 teeth 2:1:2:3

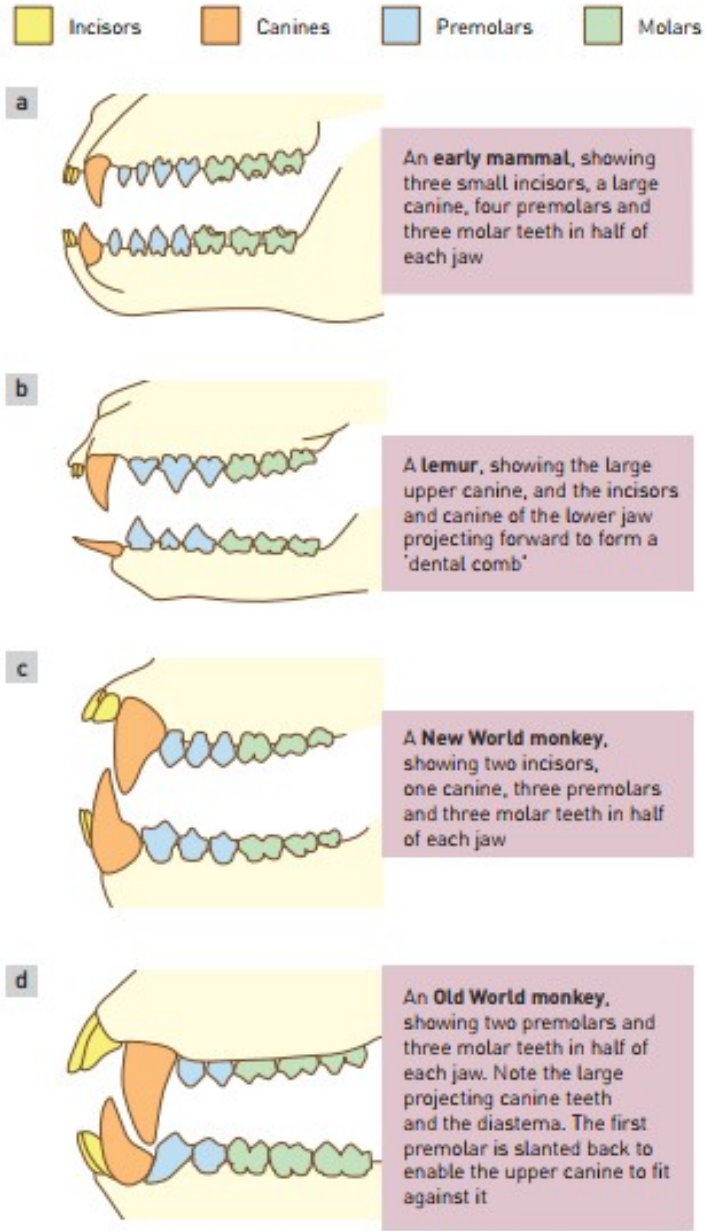


Figure 17.10 The dentition of **a** an early mammal, **b** a lemur, **c** a New World monkey and **d** an Old World monkey

Dentition (Teeth)



- Variability and specialisation in tooth form
- Lemurs and Lorises: lower incisors – dental comb:
  - Grooming
- Old World Monkeys: large, sharply pointed canines
  - Longer than surrounding teeth
  - Diastema on top jaw to fit lower canine when mouth shut
  - Lower premolar slanted back with sharp edge to sharpen upper canine



Dentition (Teeth)

- Molar teeth show little change from early mammals
  - Generalised diet, generalised teeth
  - Three cusp molar in early mammals
  - Four cusp molar in Old World monkeys
  - Apes and humans: 5 cusp molar – evolved due to fruit in diet?

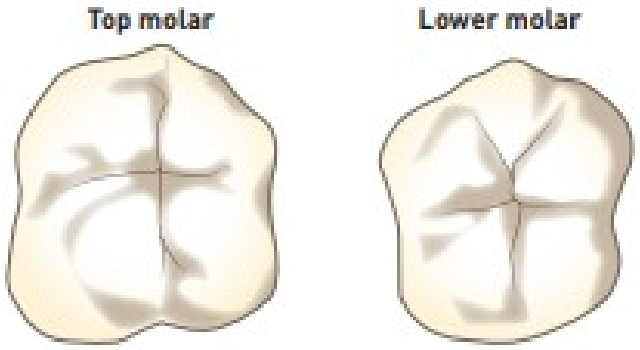


Figure 17.12 The difference in structure between a four-cusp and a five-cusp tooth. The 'valleys' between the four cusps of the top tooth form a '+' pattern, while the 'valleys' between the five cusps of the lower tooth form a 'Y'.

Dentition (Teeth)

- Fossil teeth often found due to hard enamel preserving them
- Can be used to indicate diet but caution as some adaptations are not due to diet:
  - Dental combs eg lemurs and lorises
  - Large canines in Old World Monkeys – defence and social disply

Vision

- Primates have evolved emphasis on vision
  - Due to arboreal lifestyle (in trees)
  - Decreased emphasis on olfaction (smell)
- Emphasis on vision as an advantage has led to change in skull shape:
  - Nose and snout smaller and flatter
  - Brain case larger
  - Less use of nose and snout to sense environment, greater use of eyes
    - Compare Lemurs who use snout to investigate an object vs apes who use hands and eyes
  - Movement of eyes to face forward
    - Allows stereoscopic (3D) vision, accurate distance perception essential when leaping between branches
    - Field of view narrower – could be disadvantage re: predators – compensated by having mobile head and neck – can “look around”
    - Bony socket protects eyes
  - Retina – rods and cones
    - Rods important for vision in dim light
    - Cones- visual discrimination and colour vision
  - Neural connections between eye and brain also improved
    - Acute vision
    - Coordination between eyes.

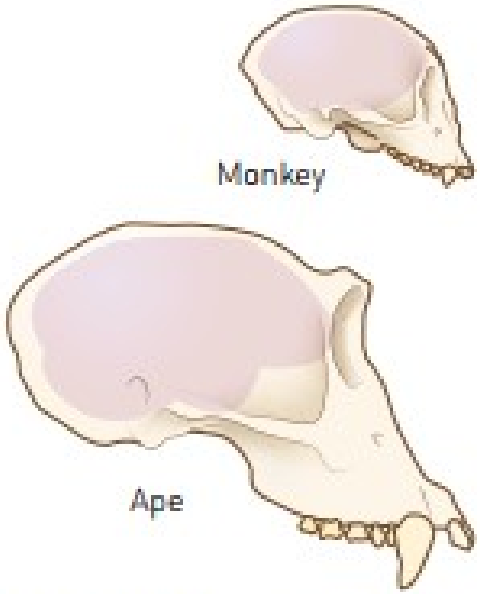


Figure 17.13 Comparison of the size of the brains and skulls of a monkey and an ape (both skulls are drawn to the same scale)

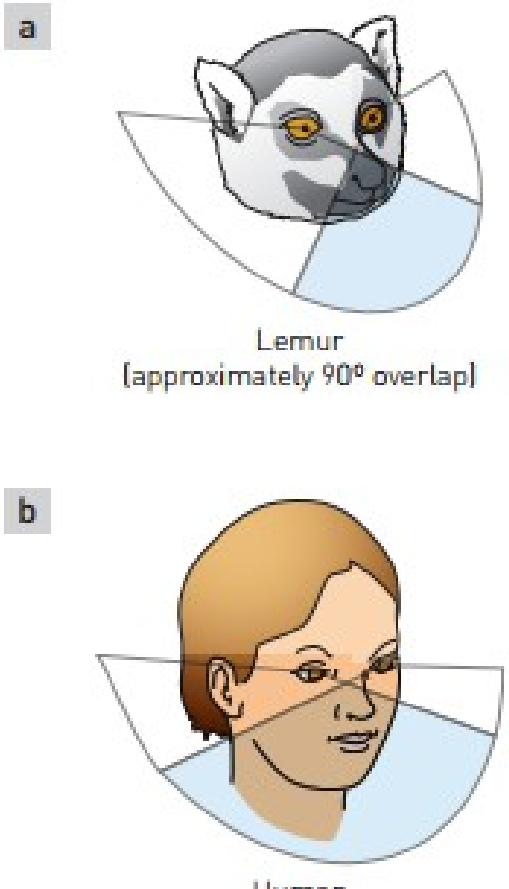


Figure 17.14 Fields of view of **a** a lemur and **b** a human. As the eyes become more forward facing, the total field of view decreases but the degree of overlap of the fields of the two eyes increases. The greater the degree of overlap, the better the stereoscopic vision.

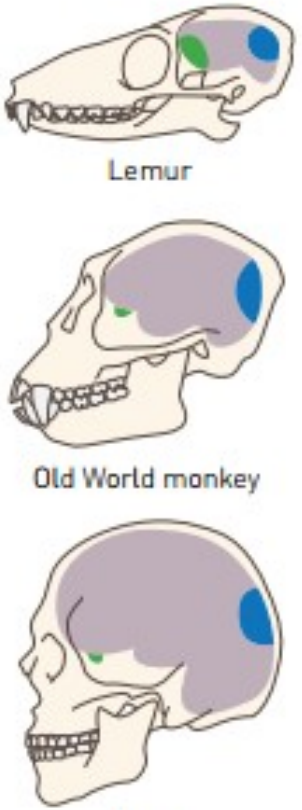


Figure 17.16 Relative sizes of the centres for olfaction (green) and vision (blue) in the brains of a lemur, an Old World monkey and a human

Vision

- Can see evolution of eye socket in comparing lemurs and apes:

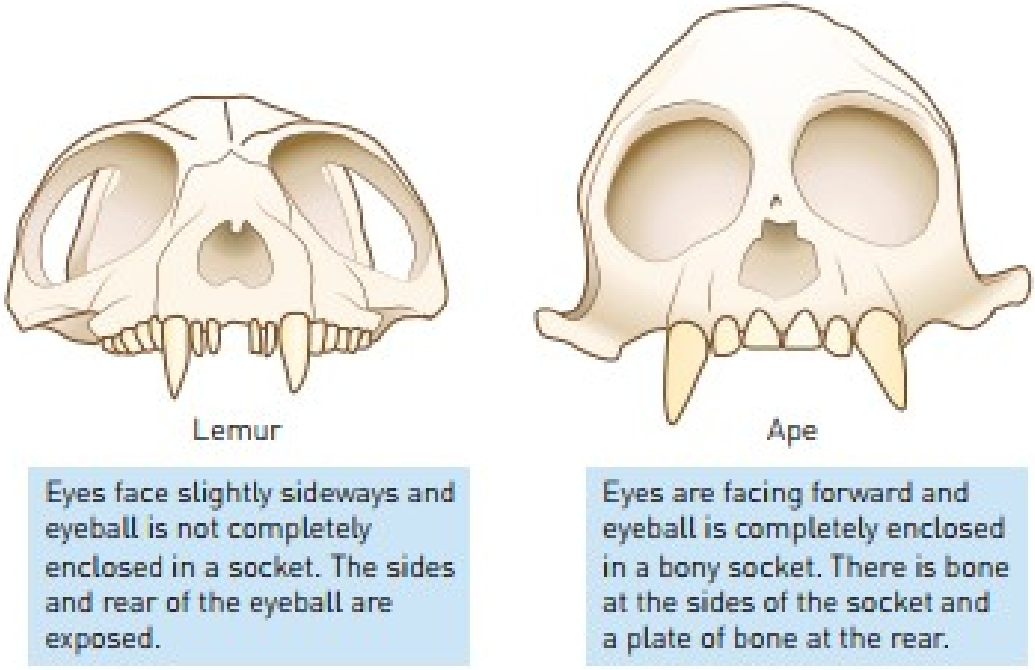


Figure 17.15 The change in the shape of the eye sockets for a lemur and an ape

- Brain regions larger for vision, smaller for olfaction as evolution moves closer to humans

Cerebral Cortex Size

- Progressive increase in cerebrum as evolution closer to human
- Region of brain concerned with “higher-order” functions
  - Reasoning
  - Memory
  - Manipulation and construction
  - Vision
  - Language
- Noticeable trend from lemurs to monkeys to humans:

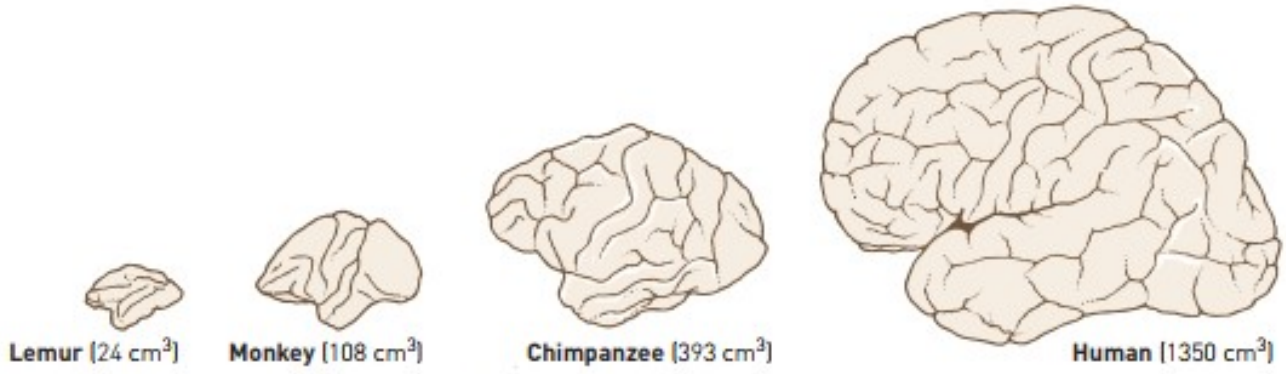


Figure 17.17 The increase in size and complexity of the cerebrum in various primates (drawn to scale)

Cerebral Cortex Size

- Large cortex compared to body size probably an advantage in arboreal environment:
  - Accurate visual and tactile perception
  - Coordination and integration of stimulus and response
  - Vision produces large amount of info that must be processed and stored
- Cortex increasingly convoluted (folded)
  - Increases surface area by 50% in humans, compared with no folds
- Increased cerebral cortex size has influenced how primates live
  - Nomadic movement
  - Food location
  - Tool construction and use
  - Variety of behaviour responses – behavioural flexibility
  - Ability to reason and problem solve
- Skull size more large and rounded in humans due to larger brain size.

Gestation and Parental Care

- Evolutionary trend in reproductive physiology and behaviour
- Generally single offspring at one time
  - Adaptation to arboreal life?
- Long period of growth and maturation with high parental care
  - Due to increased brain and skull size – need to birth before baby's head gets too big
- Primates are placental mammals
  - Offspring develop in uterus, receive nourishment from mother's blood stream via the placenta
  - Apes and humans – more efficient placenta
- Gestation (pregnancy) is long in primates
  - Tarsier: 1 offspring, 6 month gestation
  - Rodent: 6+ offspring, 3 week gestation

Gestation and Parental Care

- Long period of parental care
  - Increases in length as more closely related to humans
  - Lemurs: young weaned at 5 months
  - Apes: young feed from mother for 3-4 years, need protection and guidance for 6+ years
- Associated delay in maturation
  - Sexual maturity later in apes and humans than lorises and monkeys
  - Period of learning extended as a result
  - Allows ideas and skills to be passed on
  - Reduces number of offspring per lifecycle – time and effort must be invested in care as a trade-off

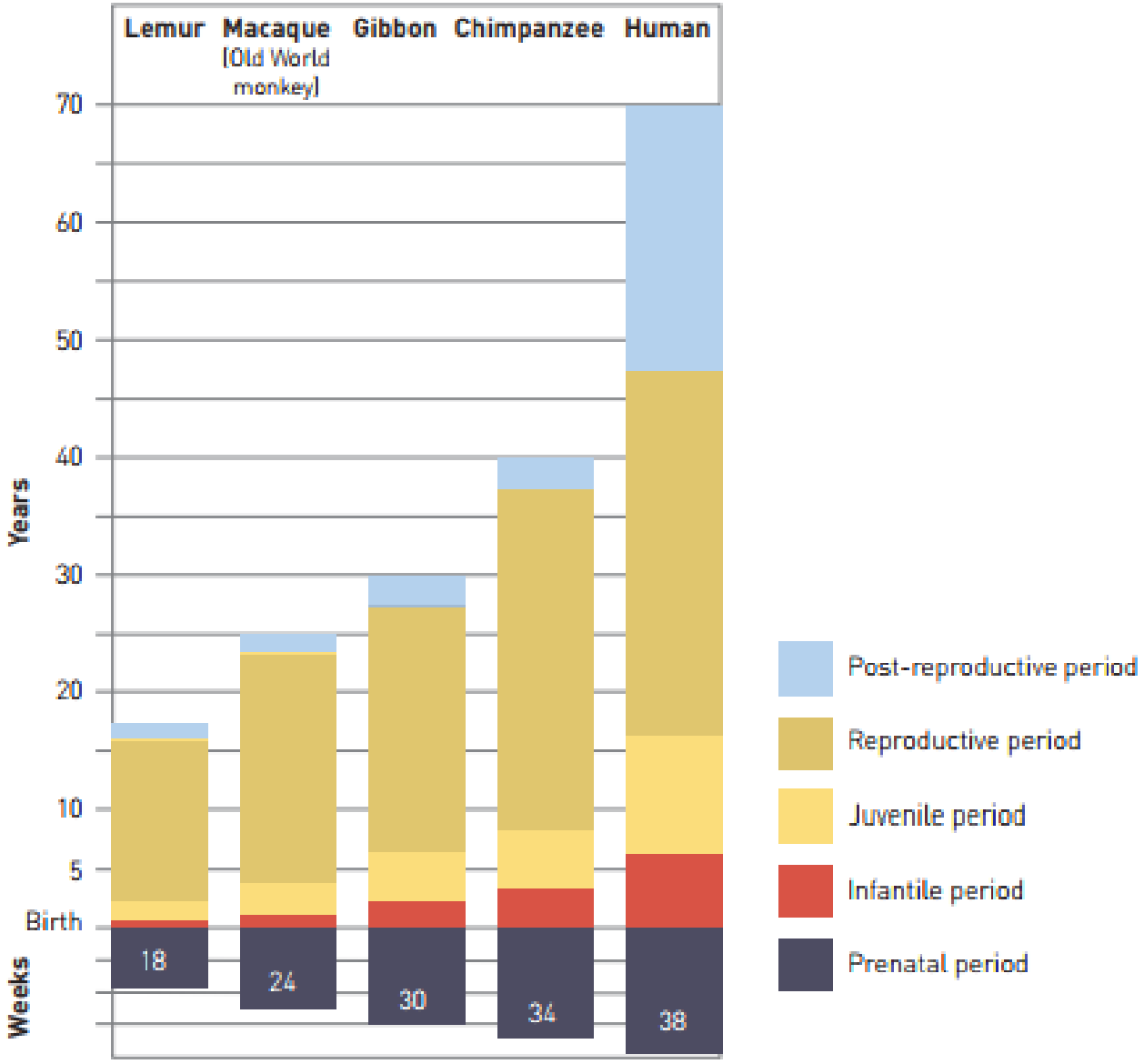


Figure 17.18 The increase in female developmental and reproductive periods with progression from the lemurs to humans

Table 17.3 A summary of the evolutionary trends that occur in the order Primates

Characteristic		Trend
Digits	Mobility	Increasing mobility and ability to move digits independently of one another
	Opposability	First digit opposable and increasing length results in increased effectiveness of opposability
	Claws/nails	Primitive primates retain claws on some digits; higher primates have nails on all digits.
Dentition		36 teeth in lemurs, lorises and New World monkeys; 32 in Old World monkeys, apes and humans. Monkeys and apes – large projecting canines with diastema 4-cusped molars in monkeys; 5 cusps in apes and humans
Smell		Sense of smell reduced with gradual reduction in length of the snout
Vision	Eyes	Increasing efficiency in vision Eyes becoming gradually more forward facing, to give stereoscopic vision
	Eye socket	Eyes gradually becoming enclosed in a protective bony socket
	Visual area of brain	Increasing proportion of the cerebrum devoted to vision
Brain	Size	Increasing size of brain relative to size of body
	Convolutions	Gradual increase in the number of folds in the surface of the cerebrum
	Cerebral cortex	Cerebral cortex making up an increasingly large proportion of the brain
Gestation		Increasing length of time between fertilisation and birth
Development	Dependence	Increasing length of time that the offspring are dependent on the parent/s
	Sexual maturity	Increasingly later development of sexual maturity